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What matters to graduate students? Experiences at a statistical consulting center from pre- to post-COVID-19 pandemic

Marianne Huebner[1,2] | Steven J. Pierce[2] | Andrew J. Dennhardt[3] | Hope Akaeze[4] | Nicole Jess[5] | Wenjuan Ma[2]

[1]Department of Statistics and Probability, Michigan State University, East Lansing, Michigan, USA

[2]Center for Statistical Training and Consulting, Michigan State University, East Lansing, Michigan, USA

[3]US Fish and Wildlife Service, Ventura, California, USA

[4]Community Evaluation Programs, Office for Public Engagement and Scholarship, University Outreach and Engagement, Michigan State University, East Lansing, Michigan, USA

[5]Department of Counseling, Educational Psychology, and Special Education, Michigan State University, East Lansing, Michigan, USA

Correspondence
Marianne Huebner, Department of Statistics and Probability, Michigan State University, East Lansing, MI, USA.
Email: huebner@msu.edu

The COVID-19 pandemic led to unprecedented changes in all levels of society, including the statistical consulting field. This paper focuses on the experiences of graduate student consultants and clients at our statistical consulting center (SCC) that operates all year independent of semesters. During the lockdown period, work continued without interruption and was conducted remotely, but there was a temporary reduction in utilization. Advice on statistical methods, help with data analysis and educational offerings are the main appeals to utilize SCC services. We describe our mentoring approach for graduate student research assistants (RAs) and how pandemic changes affected RAs and clients. Based on experiences during the pandemic, we offer practical suggestions for SCCs' approaches to research support, work characteristics and collaborations to improve the experiences of graduate students, both as consultants and clients. Most collaboration meetings are now virtual by request from clients. Telecommuting supports flexible personal schedules and needs. Online educational offerings provide easier access for participants and more opportunities for a wider range of topics and presenters. However, mentoring sessions for RAs are best conducted in-person, and every effort should be made to encourage in-person interactions and collaborations between staff members to advance the effectiveness of post-pandemic SCCs.

KEYWORDS
collaboration, COVID, graduate students, hybrid, mentoring, online, pandemic, statistical consulting, telecommuting

1 | INTRODUCTION

The COVID-19 pandemic brought numerous changes in higher education, moving from in-person meetings to online collaborations and remote work (Katz, 2021). Research studies with in-person interactions were paused or discontinued or shifted to remote data collection, where possible (Nomali et al., 2023), and laboratory research was disrupted (Sohrabi et al., 2021). Adverse effects on student careers during this time were reported, for example, challenges with pursuing research at home (Sifri et al., 2022), less access to faculty or difficulties with job searches (Katz, 2021). Longer term consequences on research and training programmes are unclear (Sohrabi et al., 2021). While biomedical statistical consulting units at medical centers may have seen an increase in demand due to COVID-related analyses (Lee et al., 2022), there was a reduction in the number of projects at other academic statistical consulting centers (Niu, 2023).

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Graduate students faced limited resources and heightened uncertainty navigating their programmes amidst the pandemic's disruptions. Statistical consulting units can provide guidance to graduate students at critical junctures of their academic journeys. However, experiences of graduate student consultants and clients from different research fields regarding research support, work characteristics and training in academic statistical consulting centers have not been studied.

Every statistical consulting and collaboration unit (SCC) is unique (Vance, 2015). The names vary from 'statistical consulting center' and 'data science consulting and collaboration units' to 'statistical lab' among many other terms. Most SCCs are open-ended in supporting research involving data ranging from software support to guidance on study design and guiding or executing data analysis. Some SCCs have training components offering seminars or workshops. Mentoring graduate student consultants and supporting graduate student research is a key mission for many SCCs. A thriving SCC develops partnerships and networks across the institution, for example, with multidisciplinary teams, individual departments or colleges who seek to provide their members with access to statistical expertise (Huebner et al., 2023). One such partnership may be with the Graduate School to facilitate building capacity in statistical and computational skills of graduate students working on their theses or dissertations, to provide opportunities for graduate students to become expert consultants and collaborators or to support educational events, especially for responsible conduct of research in data science.

The purpose of this paper is to describe experiences in an academic SCC related to graduate students, both as consultants and clients, to highlight their experiences and needs during the COVID-19 pandemic and to provide recommendations for future practices.

This is in the case study of Michigan State University's Center for Statistical Training and Consulting and includes experiences from RAs who were employed before and during the COVID-19 pandemic. In our SCC, full-time staff and doctoral graduate students are available for consulting and collaborations year-round. Collaborations are sought by hundreds of researchers (faculty, staff and graduate students) across the university every year.

In Section 2, we describe appeals and barriers to seeking out and utilizing statistical support by graduate students. In Section 3, we discuss onboarding and mentoring of new RAs pre COVID-19 and changes during the pandemic. Experiences of graduate students about research support, degree progress, work characteristics and collaborations are described in Section 4. Finally, we conclude with practical suggestions for an SCC to support graduate students.

2 | ORGANIZATION AND UTILIZATION OF THE SCC

The mission of the Center for Statistical Training and Consulting at Michigan State University is to improve quality and impact of research, to help scholars and their research programmes thrive and to increase scientific rigour and reproducibility. The SCC provides expertise and guidance on study design, statistical methods, interpretation of results, data management, conducting statistical analyses and writing reproducible reports. While the Department of Statistics and Probability started a statistical consulting service in the 1980s, resulting from these efforts the SCC was established in 2004 with an administrative home in the Office of Research and Innovation (Gilliland & Stapleton, 2013). SCC staff consists of full-time statisticians (employed in a continuing system rather than fixed term), doctoral student RAs and administrative support staff. The services are usually free for individuals affiliated with sponsoring units and fee-for-service for others. The SCC partners with The Graduate School so that SCC statisticians can provide free support to graduate students working on their theses and dissertations to conduct appropriate statistical analyses, interpret the results correctly and help them learn about important statistical methods and tools to advance their professional development. Graduate students seeking statistical support for their research projects account for 35% to 40% of SCC's cases on average.

Graduate students have access to statistical support with a set number of hours per year as well as opportunities to attend seminars or workshops offered by the SCC. The number of available hours to student clients differs by college. Fees apply for continued support beyond those hours. Some colleges sponsor the SCC with an annual fixed contribution or with a percent full-time equivalent (FTE) and thus provide free access to statistical services for their members. Information about SCC services is made known through communication channels in colleges, graduate school newsletters, SCC staff participating in resource fairs or presenting at student seminars or through the SCC website and SCC emails. A communication manager to help with advertising and effective dissemination of events, opportunities and services is not available at this time.

2.1 | Appeals and barriers to utilizing the services of the SCC

In Fall 2021, faculty and graduate students were asked to provide feedback in an anonymous survey conducted to understand their quantitative needs that the SCC could help with. While this survey was given during the pandemic, the questions were not specific to this period. Results were reported in summary form on the SCC website and reproduced here (https://cstat.msu.edu/about). Survey questions included 'What attracts you to work with [SCC] statisticians or utilize [SCC] services?' and 'What barriers make it harder for you to work with [SCC] statisticians or utilize [SCC] services?' with a Likert scale from strongly agree to strongly disagree. The response options for the question 'What are your statistical training needs?' included topics listed on the SCC website.

The overwhelming majority of graduate student survey respondents (Table 1) appreciate advice on statistical methods (81.1%), SCC staff help conducting the analysis (70.1%) and educational offerings (80.3%). Graduate students would like to see seminars or workshops on statistical methods (78.7%) followed by training on statistical software (58.3%) and data visualization (59.1%). The largest barrier noted by graduate student

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TABLE 1 Appeals and barriers to utilizing the SCC services and desired seminars or workshops for graduate students ($N = 127$ respondents)

Appeals[a]		Desired educational offerings[b]		Barriers[a]	
Educational offerings	80.3%				
Advising on statistical methods	81.1%	Statistical methods	78.7%	Lack of information about SCC	66.9%
Conducting data analysis	70.1%	Statistical software	58.3%	Financial constraints	42.5%
Study design	55.1%	Study design	41.7%	Lack of time	41.7%
Collaborating on manuscripts	45.7%	Data visualization	59.1%	Expertise not available in SCC	19.7%
Collaborating on grant proposals	40.1%	Data management	45.7%		

[a]Proportion of graduate students who answered 'strongly agree' or 'agree'.
[b]Proportion of graduate students selecting this option.

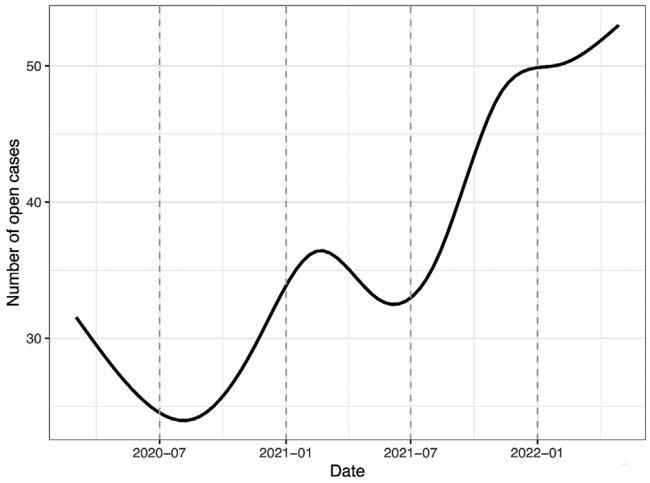


FIGURE 1 Time series of daily open cases from graduate student clients from 2020-03-01 to 2022-03-01. Dashed vertical lines indicate the start of the fiscal year on 1 July and the calendar year on 1 January, respectively.

respondents is lack of information about the SCC (66.9%). Since not all colleges are SCC sponsors providing extended access to SCC services, financial constraints can be a barrier (42.5%). When expertise is not available in the SCC, every effort is made to refer to appropriate experts or point to relevant publications.

2.2 | Pandemic impact on utilization of SCC services

SCC staff members continued to work remotely following the lockdown on 12 March 2020. The number of open cases from graduate student research projects decreased (Figure 1). SCC's policy is to close cases when there is a long halt, and clients are invited to submit new cases when they are ready to restart, for more transparent consultant case load and limiting potential scope creep. Brief downturns are expected when students graduate and leave the university at the end of Spring semester, but the decline was more pronounced in Spring 2020 compared to the same period in 2021. By the end of Fall 2020, the case counts had recovered and have since exceeded prior counts. This is in part due to new support from the Graduate School that started in May 2020 and due to new partnerships with the medical colleges that grew over time. This additional support facilitated graduate students' research during the lockdown as is illustrated by the fact that the proportion of cases from graduate students increased compared to all open cases even during the lockdown (Figure 2).

3 | HELPING STUDENTS BECOME EXPERT CONSULTANTS AND COLLABORATORS

3.1 | Onboarding and mentoring approaches of RAs

New RAs have orientation meetings followed by on-the-job training. The orientation meetings are led by the associate director, who is the supervisor of RAs, and are split into four core sessions totaling 6 h (Table 2). The core sessions cover the SCC's mission and core values, the RA's role, ethical guidelines (American Statistical Association, ASA, 2022), applicable policies and expectations (e.g., expected weekly effort, work schedules,

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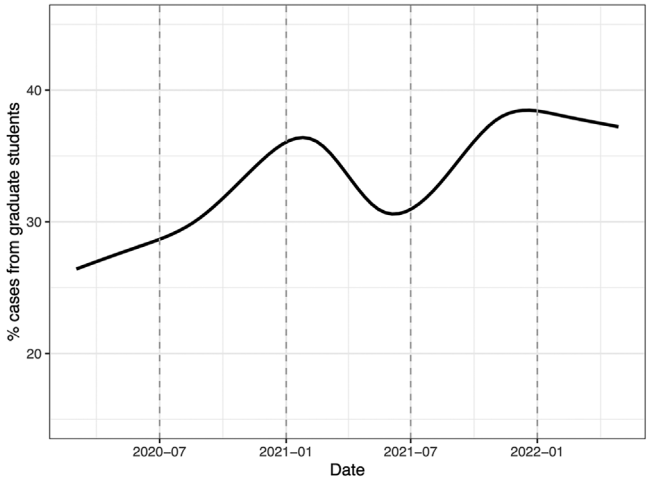


FIGURE 2 Proportion of open cases from graduate student clients of all open cases from 2020-03-01 to 2022-03-01. Dashed vertical lines indicate the start of the fiscal year on 1 July and the calendar year on 1 January, respectively.

TABLE 2 Orientation schedule for new graduate student consultants

Session	Topics	Supporting materials
Week 1		
Core 1	SCC mission and core values	SCC employee handbook
	SCC organization	SCC employee handbook
	Ethics	ASA ethics guidelines
	Expectations of research assistants	University policies; SCC employee handbook
Core 2	SCC project database (web application)	Database demonstration Online user manual for database
	Communication tools	Microsoft Teams, Zoom
	Software for reproducible research	Handout on software (Git, R, RStudio, Pandoc, Quarto, and TinyTeX); resources on using Git
Core 3	Planning and running a meeting	Papers on statistical consulting
	Organizing information and notes	SCC employee handbook
	Asking effective questions	Papers on communication
	Project and time management	
Week 2		
	On the job training starts	Methods relevant to case; support by another consultant
	Informal meeting with the director	
Week 3		
Core 4	Bids and negotiating fee-for-service	Instruction document; bid template files; SCC project database;

Abbreviations: ASA, American Statistical Association; SCC, statistical consulting center.

consulting procedures, storing project materials, tracking time and authorship) and recommended readings about statistical consulting and communication (Derr, 1999; Kenett & Thyregod, 2006; Kirk, 1991; Stoddard, 2012; Vance et al., 2022; Yu et al., 2018; Zahn & Isenberg, 1983). Project- and time-management tools and associated strategies are discussed since RAs are managing several projects concurrently. Graduate students are required by the Graduate School to obtain annual training on responsible conduct of research, so RAs in the SCC have some familiarity with human and animal subjects' protocols, authorship, data management and collaborative research.

Following orientation, new RAs learn on the job as 'primary consultant' on three to five client cases, with support from a more experienced 'secondary consultant'. Observing other consultants and having secondary consultants for their first projects are effective models for preparing RAs utilized by the SCC (Kaseda et al., 2022). The secondary consultants meet with the new RA to plan for upcoming client meetings and identify key methodological issues that may be relevant. In client meetings, the secondary consultants observe performance, help by asking thought-provoking questions and offer strategic suggestions, but the new RA is encouraged to lead the meeting. Debriefing sessions following the meeting are used to reflect on how meetings went, provide feedback, teach communication and consulting skills and plan next steps for the project. These forward-looking discussions often focus on crucial research design issues, analysis options, constraints on potential solutions and relevant articles

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or methodology resources. Pairing each RA with several different colleagues over time exposes them to a variety of consulting styles and builds relationships within our team. Video consulting assessment has not been used on the mentoring session due to time constraints but can be useful tools in training consultants (Sima et al., 2020).

As new RAs' skills and confidence grow, there is less need for secondary consultants, and they begin handling their own caseload solo, and their caseloads grow. Caseloads can vary considerably due to variation in both the scope of work and frequency of client contact. This is a consequence of clients' varying needs, preferences, skills and schedules. Most of our RAs can handle a caseload of 10–15 simultaneous cases after about 9 months of employment. Mentoring continues as needed. This may take 1 to 3 months depending on the experience and aptitude of the RAs. RAs are encouraged to request advice or assistance, from their supervisor or other SCC members of their choice.

The weekly staff meetings provide additional learning opportunities for new and continuing RAs. They are asked to summarize a case they are working on and the approach they are taking, followed by a team discussion about relevant issues, alternative methods and offering pointers towards resources such as methodology literature. RAs are encouraged to participate in discussions of other consultants' cases and may request a discussion of a particular case they have questions about. These activities help consultants hone their communication and problem-solving skills and expand their knowledge of applied statistics.

3.2 | Pandemic impact on onboarding and mentoring of RAs

Organizational changes during the COVID-19 pandemic included conducting weekly staff meetings online and mentoring and onboarding new staff such as graduate student RAs via videoconferencing instead of in-person meetings. The SCC needed more computing equipment for home offices and staff expanded their skills regarding effective use of technology.

Our pre-pandemic process for training and mentoring RAs relied heavily on in-person contact. Physical proximity in the office facilitated many informal mentoring exchanges between the supervisor and the RAs, peer support and camaraderie among the RAs. During and after the COVID-19 pandemic, most newly hired RAs were trained remotely. While the mentoring topics did not change, some approaches to disseminating specific skills and strategies to RAs were constrained. For example, statistical approaches are often readily taught sketching conceptual models, listing relevant variables or writing equations on boards; our in-person staff meetings thrived with such tried-and-true tools. When training new SCC hires remotely, this teaching strategy was more difficult because digital whiteboards were unfamiliar, drawing with a mouse is often clumsier than with a physical pen and limited screen space makes it hard to see the whiteboard and other meeting participants at the same time. Mentoring did not include topics on effective videoconferencing other than providing feedback on audio quality or lighting, and SCC staff learned skills on their own and shared experiences.

The vastly increased use of remote work induced by the pandemic raised some policy issues we had not dealt with before. Regarding the geographic location, we settled on allowing RAs to work remotely from other places within the state temporarily and initially made limited exceptions for RAs located elsewhere in the United States, since some students moved home. The latter poses a barrier partly because time zone differences make scheduling more difficult or results in mix-ups, and distance sometimes adds lag to network connections. Concerns about export control regulations led us to prohibit working from outside the United States. During the pandemic, university policies were developed or revised and covered rules about remote work.

4 | SCC ENVIRONMENT AND RESEARCH SUPPORT FOR GRADUATE STUDENTS

A collaborative and respectful environment is of utmost importance for a thriving SCC in any circumstances. Graduate students benefit not only from research support but also from career support. This applies both to RAs and clients. Graduate student clients learn to apply appropriate statistical methods for their research projects, discover new approaches and get support to negotiate such approaches with their advisors and committees. RAs broaden their repertoire of statistical methods, explore statistical approaches for their own research projects, build networks and improve communication skills. They learn to manage multiple projects and hone their organizational and note taking skills. These skills transfer when they prepare for job applications and interviews. Such support is independent of the COVID-19 pandemic, but collaborations and work characteristics changed, and new community norms emerged.

4.1 | Research support and exploring statistical methods

Graduate students seek statistical support from the SCC for their research projects. More than 80% of survey participants responded that advice on statistical methods is the main appeal of the SCC (Table 1). Clients are matched with consultants who explain options for study designs and statistical analysis approaches.

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RAs experience a dual role in supporting their clients' research as well as benefiting from research support for themselves provided by the SCC. The varied nature of research projects undertaken by their clients across different fields exposes RAs to new methods that they may not typically encounter in their respective academic programmes. It also requires them to dig deeper into the application of methods they learned previously. This interdisciplinary exposure is instrumental in cultivating a nuanced understanding of research methodologies and allows RAs to expand their methodological toolkits. Functionally, working as an SCC RA accelerates accumulation of experience with diverse research designs and statistical methods because of the varied caseload they support.

The SCC fosters an environment of intellectual growth, empowering RAs with the freedom to explore and develop new areas of interest. As RAs expand their repertoire of statistical methodologies through their work at the SCC, they not only apply these newly acquired methods to enhance their own research but also disseminate knowledge by sharing their expertise with graduate student clients.

> I remember one of my clients jokingly pleaded with me to hold off my graduation till she graduated so I could help her navigate the issues she was having with her dissertation. According to her, she was not getting sufficient methodology guidance with her advisor but her consultations with the [SCC] was helping to fill that gap.

Graduate student clients may adopt newly learned statistical methods in their current and subsequent research projects and may introduce these methods in their fields of study.

> After I successfully changed my client's methodological approach, he came back a few months later with a new project. He wanted me to help review his R script and address reviewers' comments for the tutorial paper he wrote applying the approach in his field. I was impressed by how much time he had devoted to studying this method and discussing its relevance for his field of study. I provided the requested help, and the paper was later published.

By responding to reviewers' comments on conference and journal articles, RAs learn about pitfalls in academic writing while honing their own research communication skills.

It takes time to get an adequate understanding of the client's research objectives, context, study design and data. Initial meetings that include the graduate student and their advisor can help speed up this process. Often issues must be addressed above and beyond what the student initially identified. There may be multiple options to discuss with their comparative merits and flaws. Once a statistical analysis plan has been identified, continued help may be needed to implement such a plan. Suggesting a plan that the student cannot execute leads to frustration, anxiety and dissatisfaction. We suggest resources to facilitate building the student's knowledge and capacity to apply it. Advice on structuring the data and learning its properties in a systematic manner should accompany guidance regarding the statistical models. Such tasks may need to be completed before further meetings to inform the client and the consultant before deciding on next steps. At the same time, this process often requires the RA to do background reading and hone their own capacity and coaching skills. They guide the client to compose a coherent plan on how to solve the problem or think about ways to deal with potential obstacles. The time for robust statistical analysis and data visualization is often overlooked as a cost in grant projects. Students have shared their concern of needing subsidization beyond the available free hours of SCCs support. Such financial constraints can be a substantial barrier to utilizing SCCs services (Table 1).

4.2 | Navigating challenges with degree progress

Graduate students feel they need to make independent progress and be prepared to defend the decisions without the benefit of informal check-ins with their committee members to discuss ideas before proceeding. The SCC creates a safe space for exploring analysis ideas in consulting meetings. Explaining options for study designs and statistical analysis approaches is one of the key tasks for RAs. RAs are in a unique position to observe the dissertation process through the experiences of their graduate student clients, providing them with valuable foresight and knowledge as they work on their own degrees. RAs experience that the consulting they provide and mentoring they receive at the SCC expands their knowledge of statistical tools that may creatively reshape and enhance research outcomes for their dissertations and equips them to engage more productively with their advisors.

Sometimes students may mention conflicts with an advisor or committee, including lack of availability to give advice or lack of awareness of the work the student has already completed. This hinders students from attaining their goals or makes them feel less valued or respected. Lack of interactions negatively impact research outcomes and timelines associated with completing essential tasks. RAs may be perceived as confidants, and student clients may describe these and related challenges, especially discord concerning analytical approaches to them. RAs are encouraged to discuss cases to get advice and input from the team in the weekly staff meetings or from SCC supervisors, who can step in to join meetings with clients as needed. The SCC invited the university ombudsperson to a staff meeting to help with awareness and learn about resources that can then be passed along to students.

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During the pandemic, students encountered disruptions to their research projects. Constraints of the pandemic necessitated changes to research plans such as adjustments to timelines or data collection strategies. Some planned statistical analyses needed to be adjusted to account for pandemic effects. Students' research processes were further disrupted by less frequent interactions with mentors and colleagues and general declines in productivity. RAs found themselves in the position of assisting their clients with navigating these new barriers to their research, while facing similar disruptions to their own research.

4.3 | Pandemic impact on collaborations and videoconferencing

The abrupt shift from in-person to remote consulting during the lockdown presented substantial challenges. Not all consultants and clients possessed a home office environment conducive to regular virtual meetings. This led to initial delays and difficulties as individuals set up the necessary equipment and familiarized themselves with video conferencing tools. Unreliable internet connections occasionally interrupted video or audio streams, forcing meeting participants to repeat themselves, exit the meeting and reconnect or even postpone meetings.

The SCC started a lending programme of laptops for RAs. The continuity and support provided by the SCC played a vital role in helping RAs through this period of transition, although the success of the transition to online meetings with clients varied.

> Videoconferencing with clients were not as smooth because clients were at varying levels of technological familiarity.

Despite the initial hurdles, consultants found some positive aspects in switching to remote meetings. The transition demanded creative solutions to maintain the quality and effectiveness of client engagements, not only navigating the obstacles imposed by the pandemic but also spurring the adoption of tools and techniques that are useful for both remote and in-person meetings going forward.

> Initially, in-person meetings with my client involved walking her through how to use SPSS for her analysis using verbal descriptions and on-screen guidance, with the client taking notes on the process. The transition from in-person to remote sessions posed some challenges to this method (for instance, I could not physically point to where to click on her screen), so we had to adjust our approach. Using Zoom, I could request control of the client's screen, enabling real-time demonstrations while the client recorded the session. This adaptation proved much more efficient, allowing me to provide explanations and address questions throughout the process, while providing the client with a valuable reference tool for future use.

RAs found camaraderie and support from both SCC colleagues and clients as they collectively adapted to the unique challenges of conducting research during the pandemic.

> After a few Zoom meetings, we established a rapport and, given we were both working on our dissertations then, we turned the Zoom meeting to a study room, with each person working on her own thesis. With this arrangement, she could unmute herself from time to time to ask for help on her work. The lockdown and use of Zoom had created a less formal connection between us that went beyond the [SCC]. To date, this client, now an assistant professor in a university, still joins me on Zoom from her office so provide each other with virtual company while working on our separate job tasks.

The virtual consultation during and post COVID pandemic has made consulting meetings more accessible for busy advisors and allowed them to attend the meetings together with their student more frequently. This not only empowered the student client but also provided a professional development platform for the RAs. RAs can learn from the domain expertise of the advisors, which may lead to future collaborations beyond the SCC activities.

> The mentoring and consulting sessions not only improved my statistical skills but also opened several networking opportunities for me. Through my consulting sessions, I learned the expertise of the students and faculty members I worked with and found some that were relevant to my own dissertation work. Eventually, my own dissertation committee included a previous faculty consulting client, and an [SCC] alum. After I graduated, the same faculty member invited me to serve as a dissertation committee member for one of her graduate students who was a RA at the [SCC].

However, explaining problems can take longer during a virtual meeting than during an in-person meeting for both clients and consultants.

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4.4 | Work characteristics and collaborations

The SCC fosters a respectful and inclusive work environment. Our core values were developed with input from all staff members and graduate students. The collaborative environment enables RAs to engage in meaningful discussions about their consulting projects as well as their own research. These discussions are greatly enriched by the diverse academic backgrounds and perspectives of their fellow consultants. The diversity of client projects and associated programmes begets diversity of client-RA interactions, both in terms of scientific topics and disciplines as well as socioeconomic backgrounds, thus enriching the collaborative research experience and encouraging professional growth.

Pre-pandemic video consulting was sporadic and initiated due to clients who could not easily attend meetings on campus. Such cases often depended on constraints related to the clients' resources, time and abilities to travel from one end of a large campus to another, where the SCC was housed. A few instances relied on video consulting to continue client-consultant collaborations during travel. Video consulting was not a tool that our team used regularly before the pandemic. Switching to screen-sharing interactions and associated lessons and materials was not seamless for most consultants and their clients. Interactions between staff and clients became less personable, especially when client or consultant participants chose to keep their videoconferencing cameras turned off. With time and practice, amidst a growing demand for consulting services, client-consultant pairs adapted and honed their technology skills and became more patient and understanding. There were fewer opportunities to informally interact with the other RAs for peer support and camaraderie between meetings.

> The switch to fully remote staff meetings was smoother for me than it was for my other job commitments during the lockdown. Remote staff meetings were fine at first because I thought it was just going to last for a short while, but, with time, the impersonal nature of this form of meeting started to tell on me.

To help mitigate the lack of interaction, new informal, but virtual, monthly meetings between the SCC director and RAs were introduced. They provided opportunities to chat about client cases, their own research, personal experiences or anything else they wanted to discuss.

Conducting meetings online involves screen sharing and using a whiteboard, but this approach was neither seamless nor natural, particularly compared to in-person instruction and use of a board to sketch out key concepts. Instruction through videoconferencing can make the client-consultant experience and relationship more difficult to navigate and accurately gauge the understanding of clients in need of help and virtual tools often lengthened the time for both clients and consultants to explain problems, teach new skills and enhance conceptual understandings. Anecdotally, the increased use of videoconferencing may have positively correlated with increased numbers of clients missed, postponed or cancelled meetings. Such occurrences seemed related to clients lacking traditional time and resource investments such as leaving home to travel to work and thus commuting between academic buildings to accomplish various tasks. However, the convenience of online scheduling and virtual instruction enabled faster project-turnaround times, recordings of instructional sessions for subsequent client and consultant review and enhanced learning, and the effective use of available technology improved over time as described in the following two experiences.

> The flexibility in scheduling meetings online enabled a more efficient workflow with clients. Before COVID, when my client had a quick question or issue that impeded their progress, they typically saved it for our next meeting or I would try to assist them via email, which can be challenging. It was usually impractical to arrange a short meeting the same day due to the need for the client to travel across campus to [SCC]. However, after transitioning to online meetings, we were often able to schedule a brief session within a few hours to address the issue, enabling the client to continue to make progress on their work with minimal disruptions.

The convenience of remote work was appreciated.

> I could attend meetings with my [SCC] clients, work on my dissertation, and meet the responsibilities of my new job all without leaving my desk.

Now online consulting meetings are the most popular choice. Our intake form for new projects includes an item asking about preferences for meeting modality. More than 50% request online meetings (a higher proportion of 70% coinciding with graduation deadlines), 6% in-person and the remaining do not indicate a preference. However, when an in-person meeting is offered, they invariably also choose the online option. This is related to the desire of clients with demanding daily work schedules to minimize their time away from labs or clinics or reduce productivity costs spent travelling or finding parking.

The mentoring and consulting format at the SCC provide RAs with many career and personal development opportunities including workflow and project management skills. For some projects, RAs negotiate fee-for-service bids. To be successful in the bid process, RAs get instructions on how to categorize the work required for a case and estimate the time effort needed for each of these categories. They agree on a statement of work with their clients and negotiate the associated cost estimates (Peterson et al., 2022). These are important skills for future grant writing and preparing for careers.

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4.5 | Educational opportunities

Seminars and workshops moved online and thus were more accessible across the university due to the convenience of attending remotely. SCC seminar offerings were expanded due to the overwhelming interest expressed in the survey (Table 2) and now included presentations by internationally renowned statisticians, as well as additional summer seminar series with specific themes (Reporting Guidelines, Data Ethics, Open Science, Data Visualization), and introducing additional topics related to responsible conduct of research (RCR). The attendance in SCC seminars doubled compared to pre-pandemic years, with a 58% increase in participation from the natural sciences, engineering and agriculture; a 25% increase in social sciences, education and communication; and 84% increase from the medical colleges. It also allowed participation of international students who were unable to travel or advisees of the presenters at other universities. Despite the interest in educational offerings on statistical methods or software (Table 2), the number of 3-h workshops was greatly reduced due to the time commitment for SCC staff in preparing materials and lack of interest with respect to number of participants willing or able to commit this time.

The RCR seminars continue to be offered with topics related to Rigor and Reproducibility and Data Management in quantitative sciences. Before receiving credit, students are required to answer questions about the seminar content via an online survey. This is also an opportunity to provide feedback and request additional topics of interest. RCR seminars are especially appreciated since they give more opportunities throughout the year in addition to what is offered by the Graduate School and due to being more in-depth about RCR practices in the quantitative sciences as illustrated by some of the feedback. Students comment that this covers new information and that they find this more applicable to their own research than the typical topics repeated each year in RCR workshops. For example, students may not be encouraged by their research group or get the time and support to do a thorough data cleaning process and to understand data properties before rushing to sophisticated analyses. There is a lack of knowledge about implications for the interpretation of results and generalizability because of excluding data samples, overfitting, p hacking or ad hoc decisions during the statistical analysis process are some of the topics that are often not covered in their curriculum.

5 | DISCUSSION AND OUTLOOK

The COVID-19 outbreak brought many changes to the research community and the workflow of statistical consulting centers. College students struggled with social distancing (Hamlin & Barney, 2022) and researchers experienced more isolation during the COVID-19 pandemic (Katz, 2021). In a survey of chemistry departments, graduate students and postdoctoral fellows reported that the quality of collaborations decreased, resulting in less interactions, lack of communication and reduced collaborations (Sifri et al., 2022). A statistical consulting center where the main consultants are graduate students taking consulting courses during semesters reported the lowest number of cases in Fall 2021 and recovery to pre-pandemic levels by Spring 2022 (Niu, 2023). Our SCC is continuously staffed throughout the year by RAs and full-time staff. We observed a steady decrease in the number of cases by graduate students in the months following the lockdown, but the case counts had recovered by the end of Fall 2020. The importance of at least some free access to statistical consulting is evidenced by the increasing proportion of graduate student projects in the SCCs portfolio when this was made possible due to the Graduate School's support. For the SCC's sustainability and growth and for high quality research outcomes, it is essential that funding sources are made available for university-specific priorities, such as providing statistical support and training to students and faculty across the university or support for grant proposal development (Huebner et al., 2023).

New community norms for researchers emerged during the pandemic including a desire and openness for remote work and hybrid events (Katz, 2021; Niu, 2023). At our SCC, most meeting modalities are now requests for online meetings and remote collaborations. Telecommuting and hybrid staff meetings support flexible personal schedules and needs, but too much screen time can be exhausting (Trott et al., 2022).

Irrespective of the COVID era graduate students value guidance for research and degree progress, help with achieving their career goals, and respectful and inclusive work environments (Sifri et al., 2022). A statistical consultant has multiple roles: expert professional, communicator, collaborator, leader, advocate and scholar (Zapf et al., 2019). RAs learn skills to listen carefully and to ask questions, explain statistical methods and their usage and empower their clients to build their capacity (Vance & Smith, 2019). Clients gain a deeper understanding of statistical methods, appropriately interpret the results and explain and justify their choices to their advisors and committee members. RAs in the SCC fulfil dual roles of supporting graduate student clients while having to deal with their own stresses regarding completion of their degree and job searches.

Based on our experiences, we offer suggestions for SCC practices to benefit graduate student clients and consultants.

1. Environment: a respectful and inclusive work environment is of utmost importance for a thriving SCC. It provides RAs with career and personal development opportunities. An SCC can be a 'safe space' for graduate students to explore statistical methods for their research ideas. The experiences directly benefit RAs by enhancing their own pursuits of producing high-quality, collaborative research. Furthermore, a quality scholarly environment attracts clients and new opportunities for collaboration, a lifeline for great business and strong community.
2. Staff meetings: in-person staff meetings are highly valuable for collaborations, brainstorming sessions and social interactions, but hybrid meetings should also be an occasional option. Regular discussion of potential challenges in projects, including advantages and disadvantages of

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methodological choices as well as sharing new insights and discoveries, helps RAs convey similar, relevant information to their clients. It also builds their capacities as future project leaders.

3. Online meetings and remote collaborations: many graduate student clients choose this option to reduce commuting time or to be able to fit their busy schedules and disparate locations. It can result in a faster turnaround for answering questions, scheduling meetings and completing projects. The convenience of recording online meetings allows for later review and better understanding of points that may have been missed in a densely packed advising session. It is more feasible for advisors to attend the meetings. This empowers student clients by building student-advisor relationships while working towards a common goal and provides opportunities for RAs to learn from domain experts and help build collaborative interdisciplinary networks across the university. However, in-person meetings may be more productive to gauge understanding, explain multiple approaches, discuss strategies, utilize physical teaching media or progress to next steps.
4. Mentoring RAs: sessions should be conducted in-person for a richer interactive experience and to build interpersonal relationships. In-person mentoring can be augmented with quick online calls for problem solving and fast turnaround. Onboarding topics include mission and core values for the SCC, the RA's role and expectations, ethical guidelines, statistical consulting practices, digital notetaking tools and analysis software, and communication and management skills. Guidance for conducting effective online meetings is both needed and useful, exposing new staff to multiple modalities of teaching and learning. Observing other consultants through shadowing and having secondary consultants work on their first projects are effective approaches to help RAs become expert consultants and collaborators.
5. Telecommuting: RAs have for more flexibility with remote work scheduling meetings, working on client projects or making progress on their own research. The option of using office space in the SCC can provide welcome boundaries between work and personal lives. The opportunity to meet with SCC peers and share ideas and get their feedback or to browse statistics books available at the SCC can provide major benefits. Rules about telecommuting and about the expected number of in-person office days need to be carefully assessed and clearly communicated. It is important for any organization to encourage in-person collaboration to foster communities of trust and engagement.
6. Technological support: a lending programme for equipment is useful to optimize RAs set-up for videoconferencing, assures proper installation of approved analysis software and antivirus software and allows secured login to the SCC's server to run data analyses. Feedback on audio quality, lighting, presentation or background helps with a good start for building good relationships with online clients. Sharing information about SCC peers' software and hardware preferences can be useful for inspiring work efficiency.
7. Educational offerings: external speakers and local experts may be engaged as presenters to offer a broader range of topics in online seminars. Responsible conduct of research in data management, statistical analysis and reporting are of interest to many students. In-person workshops can be developed for more in-depth topics to learn skills that augment a student's curriculum and allow for interactions and individual attention. Encouraging learning new skills, best practices and recent advances benefits RAs as well as the wider university community.

In summary, this paper helps with providing a better understanding of what matters to graduate student consultants and clients at an SCC as it relates to research support, workplace characteristics and collaborations, and educational opportunities. The COVID-19 pandemic had an impact on caseload and changes in approaches to academic mentoring and collaborations. Some of the changes are quickly becoming community norms, such as receiving statistical support via videoconferencing or being able to attend online training and give presentations at considerable distances from home or office. However, onboarding and mentoring of RAs should be conducted in person, and every effort should be made to encourage in-person interactions. Environments play a key role in their professional development. Challenging, diverse, supportive and scholarly workspaces beget high-quality SCCs, staff consultants, projects, collaborations and scientific publications.

AUTHOR CONTRIBUTIONS

MH conceived of the study and wrote the first draft. SJP, AJD, HA, and NJ participated in the conceptualization and shared in writing, reviewing, and editing. WM critically reviewed and edited the final draft.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article, as no new data were created or analysed in this study.

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ORCID

Marianne Huebner https://orcid.org/0000-0002-9694-9231
Steven J. Pierce https://orcid.org/0000-0002-0679-3019
Hope Akaeze https://orcid.org/0000-0002-8877-7900
Nicole Jess https://orcid.org/0000-0002-7102-5294
Wenjuan Ma https://orcid.org/0000-0001-8637-0800

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AUTHOR BIOGRAPHIES

Dr. Marianne Huebner is the director of the Center of Statistical Training and Consulting (CSTAT) and a professor of the Department of Statistics and Probability at Michigan State University, East Lansing, MI, United States. Her research interests are initial data analysis and data quality (https://stratosida.github.io/), health outcomes and sports medicine. Her roles as director include organizing and presenting CSTAT educational events and communicating with CSTAT stakeholders.

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Dr. Steven J. Pierce is an associate director at Michigan State University's Center for Statistical Training and Consulting. Among other duties, he mentors and supervises CSTAT's graduate student research assistants and coordinates the consulting operations by assigning cases to consultants. He earned his PhD in Ecological-Community Psychology from Michigan State University. Prior to that, he spent several years working at the Michigan Public Health Institute. Expertise in applied statistics, program evaluation, data management, database design, and tools and practices for reproducible research have facilitated his research collaborations with partners from a wide variety of academic disciplines.

Dr. Andrew J. Dennhardt is a Fish and Wildlife Biologist working in the Listing and Recovery Division of an Ecological Services field office of the US Fish and Wildlife Service in Ventura, California. In this role, he reviews past and recent statuses of threatened and endangered plant and animal species as well as collaboratively implements conservation towards recovering such biota protected under the federal Endangered Species Act. With respect to research, he is interested in species distributions, individual animal movements, group demography and spatio-temporal variation in the environment, all of which collectively shapes wildlife community and population dynamics in aquatic and terrestrial systems. As a PhD student at Michigan State University, he served as a statistical consultant for faculty, research staff, post-docs and undergraduate and graduate students at the Center for Statistical Training and Consulting during 2016–2020.

Dr. Hope Akaeze oversees the data analytics team in Community Evaluation Programs, at Michigan State University. Her team is responsible for the evaluation of the characteristics, quality and impacts of Michigan-s preschool and out-of-school time programs. She is currently the co-principal investigator for the evaluation of Michigan's Great Start Readiness Program. Her research interests include the improvement and application of statistical theories and methods for creative problem solving, particularly with multilevel modelling, structural equation modelling and cognitive diagnostic measurement models. As a PhD student at Michigan State University, she provided consultations, conducted statistical analyses and collaborated on manuscripts at the Center for Statistical Training and Consulting during 2016-2020.

Nicole Jess is a PhD student in the Measurement and Quantitative Methods program at Michigan State University. Her methodological interests include instrument design, structural equation modelling and social network analysis. As a research assistant at CSTAT from 2015 to 2023, she provided data management, statistical analysis and manuscript support on research and evaluation projects spanning a wide range of disciplines, with a focus on social science, education and public health fields.

Dr. Wenjuan Ma received her PhD in the College of Communications Arts and Sciences at Michigan State University. In her role at the Center of Statistical Training and Consulting (CSTAT), she collaborates on many research projects from multiple disciplines. Her research interests are machine learning, agent-based modelling, structural equation modelling (SEM) and causal inference.

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